
Survey and Empirical Evaluation of Attacks and Defense for Robust Classification

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Abstract

Challenges to neural network classifiers these days involve corruptions at training time and at testing time. Data corruptions and class imbalance at training time can heavily degrade the generalization and robustness performance. Data corruption occurs at testing time can drastically change neural net prediction with an imperceptible amount of perturbations. As a consequence, data overfitting and adversarial examples present threats to security sensitive applications such as self-driving car. In this work, we evaluate example reweighting and adversarial training techniques to robustify a fossil classification task. Additionally, we survey recent papers about attacks and defense of adversarial examples for neural networks.

1 Introduction

Deep nets have shown to be powerful classifiers in various settings. However, data corruptions present a challenge to robustness and generalization. Corruptions at training time are often considered as learning with the presence of outliers. More generally, it can be attributed to the distribution shift between training set and test set. Class imbalance and label noise in the training set are two common problems that lead to poor generalization performance. Example reweighting algorithms are popular solutions to these problems, but most methods require careful tuning of additional hyperparameters, such as example mining schedules and regularization hyperparameters. We investigate a meta-learning algorithm [16] that learns to assign weights to training examples based on the performance on a clean validation dataset.

Corruptions can also happen in test time that act as adversarial examples to neural networks. The adversarial attacks and defense have been researched extensively these three years [20; 6; 11]. Attacks can be white-box that takes detailed model structure information with gradient or black-box that only requires model oracle calls. Associated with a perturbation set, white-box attack methods such as projected gradient descent (PGD) performs a first order update to the objective of maximizing classification loss. [11] demonstrates that this is the natural and "universal" attack with gradient information. [1] generates attacks that stably fool neural nets under natural transformations. [2] propose an network trained to generate white-box attacks. Transferable attacks are commonly used in white-box adversaries where a proxy is trained and attacked first and later adapted. An adversarial subspace is also proposed to analyze the transferability of attacks. Correspondingly, [19] and [9] use diverse ensemble to improve the robustness to adversarial attacks. Finally, [7] and [3] use only oracle calls to DNN to generate attacks efficiently.

For the empirical evaluation, we consider a fossil classification task on 224×224 grayscale images with total 6000 images and 33 classes. The job is to maximize classification accuracy on the test set, to classify the tribe of a given fossil image. Class imbalances and label noises are often present on this setting as the fossil data can be hard to collect and label correctly by humans. Moreover, test time robustness can be extremely important as a false fossil classification might mistakenly bias a biology researcher. Our experiments on reweighting examples reproduce the results on MNIST dataset and

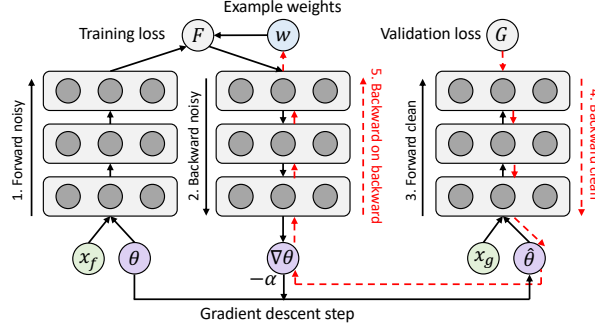


Figure 1: To get the gradients of the example weights, one needs to first unroll the gradient graph of the training batch, and then use backward-on-backward automatic differentiation to take a second order gradient pass.

show a performance improvement on fossil dataset. We also demonstrate comparable results with hyperparameters tuning. For test time corruption, we produce attacks with PGD to easily fool the trained network and show a large robustness improvement with adversarial training.

2 Meta Example Reweight Training

The key idea of meta example reweighting is that example weighting should minimize the loss of a set of unbiased clean validation examples that are consistent with the evaluation procedure. Formally, denote (x_i, y_i) as the i th input-target pair in the training set and (x_i^v, y_i^v) as the i th input-target pair in the clean validation set. Denote θ to be the neural network parameters and $f_i(\theta)$, $f_i^v(\theta)$ to be the loss on the i th training example and validation example respectively.

The goal is to learn a reweighting of the inputs to minimize a weighted loss:

$$\theta^*(w) = \arg \min_{\theta} \sum_{i=1}^N w_i f_i(\theta) \quad (1)$$

with w_i unknown upon beginning. The optimal selection of w is based on its validation performance:

$$w^* = \arg \min_{w \geq 0} \frac{1}{M} \sum_{i=1}^M w_i f_i^v(\theta^*(w)) \quad (2)$$

Consider perturbing the weighting by ϵ_i for each training example in the batch

$$\begin{aligned} f_{i,\epsilon}(\theta) &= \epsilon_i f_i(\theta) \\ \hat{\theta}_{t+1}(\epsilon) &= \theta_t - \alpha \nabla \sum_i f_{i,\epsilon}(\theta) |_{\theta=\theta_t} \end{aligned} \quad (3)$$

Then we can take a single gradient descent step on a mini-batch of validation samples with respect to ϵ_t to online approximate the minimizer

$$\epsilon_t^* = \arg \min_{\epsilon} \sum_{i=1}^M f_i^v(\theta_{t+1}(\epsilon)) \quad (4)$$

This method can be easily implemented with standard autodifferentiation package [15], as shown in the Fig. 1. Note that the meta example reweighting schedule is different from finetuning in that the clean validation acts more like a regularizer rather than a data source.

We first reproduce its result on a highly imbalanced MNIST subset containing 5000 images with 25 images of classes 4 and 4975 images of 9. The network, LeNet in this case, is able to learn something nontrivial compared with normal training. In our experiments, we use meta-reweighting to finetune the ImageNet-pretrained network on our fossil dataset. On our harder fossil dataset, shown in Table 1, there's a nontrivial improvement using example-reweighting. However, a hand-tuning

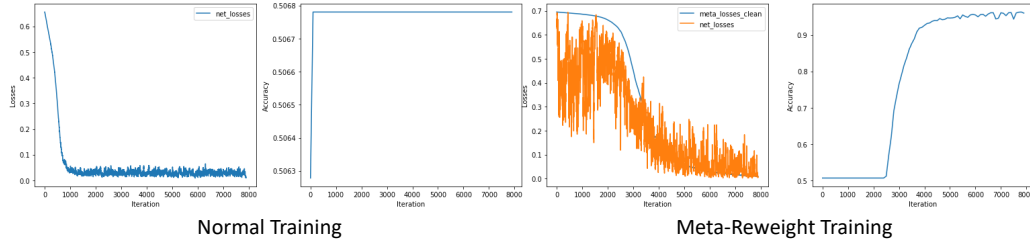


Figure 2: On the imbalanced MNIST dataset, (left) it’s not surprised that normal training on such imbalanced dataset won’t generalize to any meaningful classification. Indeed it reaches 50 percent on test set, the same as random guessing. (Right) The meta example reweighting algorithm helps the network learn to differentiate between 4 and 9 and achieve an accuracy of 98 percent. This implies the network takes advantage of the clean validation set as well as the size of the training set.

hyperparameter (momentum and weight decay) with a more advanced optimizer (ADAM) can reach comparable results. Furthermore, meta-reweighting takes up large GPU resource and time with a relatively large validation set. We were unable to finish the training experiments with VGG11 on a clean validation size 500 and a NVIDIA TITAN X.

3 Adversarial Training

For an example pair (x, y) , network function f , and loss function L , a natural attack is to simply find a perturbation which maximizes the loss of x , subject to the constraint that it lives within P_x

$$x^* = \arg \max_{x' \in P_x} L(f(x'), y) \quad (5)$$

let π be the projection onto P_x , for some step-size schedule η_t , we can define the sequence of updates $x_0 = x$, and

$$x_{t+1} = \pi(x_t + \eta_t \nabla L(f(x_{t+1}), y)) \quad (6)$$

This gradient ascent update does well in attacking normally trained neural networks. PGD can be targeted to specific class or untargeted to lower the likelihood prediction of the true class. While the input image is perturbed in imperceptible amount from human eyes (Fig. 3), a state-of-the-art neural net can be easily fooled to have near 0 classification accuracy.

A common empirical defense is called adversarial training, trying to run SGD on the robust loss of the classifier directly [11]. For an ϵ size constraint in L_p norm for the perturbation, the gradient of the robust loss of f can be written as

$$\mathbb{E}_{(x,y) \sim \mathcal{D}} [\nabla_{\theta} (\sup_{\delta \in \mathcal{P}_{p,\epsilon}} L(f(x + \delta), y))] \quad (7)$$

By Danskin’s theorem, to evaluate the gradient of the supremum of a class of functions, one should evaluate the gradient of the function in the coordinate that actually obtains the maximum. So, an iteration of the adversarial training can be run with adversarial attack, such as PGD. In a high level, the adversarial training robustifies the decision boundary of the classifier. As shown in Table 1, we evaluate the performance of various training with architecture ResNet18 [18] and VGG11 [8] on adversarial attacks [5]. The adversarial examples are generated with different constraint size for 10 steps on untargeted classes.

4 More Empirical Attacks and Defence

While PGD generates adversarial attacks to a single network efficiently, it requires first order information and is not invariant to viewpoint shifts, camera noise, and other natural transformations. This is also true for other optimization-based approaches such as Fast Gradient Sign Method or Iterative Attacks [6]. In this section, we explore a few empirical white-box attacks and black-box attacks, as well as transferability and zeroth order attack.

Evaluation of robustness to adversarial attacks using PGD						
Network	Resnet				VGG	
Stat./Alg.	Baseline	Tune.	Reweight	Adv.	Tune.	Adv.
L_2 : Clean	80.76%	84.91%	86.97%	81.06%	82.88%	78.48%
L_2 : 0.001	77.42%	85.00%	85.61%	81.06%	82.58%	78.03%
L_2 : 0.01	51.36%	74.70%	76.06%	79.39%	78.18%	76.52%
L_2 : 0.1	0%	4.85%	14.24%	51.52%	17.12%	40.61%
L_∞ : Clean	80.76%	85.91%	86.97%	84.09%	82.88%	66.67%
L_∞ : 0.00001	75.76%	84.39%	85.61%	83.79%	82.42%	63.79%
L_∞ : 0.0001	36.82%	69.39%	73.48%	80.91%	77.27%	61.52%
L_∞ : 0.001	0%	5.45%	13.48%	55.91%	18.03%	46.52%

Table 1: We evaluate the robustness of trained network on adversarial examples using PGD on L_2 and L_∞ norm constraints. Baseline denotes normal training schedule to minimize loss on training set. Tune denotes tuning on hyperparameter for training. Reweight denotes meta example reweighting method. Adv. denotes adversarial training. From the result, we see a huge improvement by adversarial training on adversarial attacks and an improvement of generalization of reweighting procedure. We also note that reweighting and larger capacity network [11] such as VGG provides some defense to adversarial attacks.

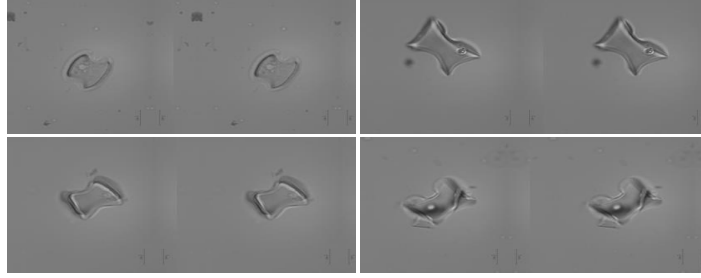


Figure 3: (Left) Original image. (Right) Example adversarial image generated PGD with perturbation constraint 0.001 in infinity norm, i.e the grayscale value of these two images do not differ by 0.001 per pixel (0.2 pixel in 256 pixel scale). Such small level of perturbation is able to fool a ResNet18 while the perturbed example looks exactly the same as original image to human eyes. For instance, on the bottom left, the left image is correctly classified as *guaduellieae*, and the right one is predicted as *arundinarieae*.

4.1 Expectation Over Transformation for Robust PGD Attack

White-box adversarial examples compute gradients from the loss function with respect to the input pixel. However, The perturbations carefully crafted by standard techniques often fail to synthesize robust adversarial examples that are relevant in real world [10], due to natural phenomena such as lighting and camera noises. [1] propose Expectation Over Transformation (EOT) algorithm to construct attacks that remain adversarial after transformation t from a chosen distribution T , as an example shown on Figure 4. Moreover, for imperceptible perturbation, EOT aims to constrain the expected effective distance between the adversarial and original inputs,

$$\delta = \mathbb{E}_{t \sim T}[d(t(x'), t(x))] \quad (8)$$

For instance, for a rendered 3D image, this would minimize the visual difference instead of the difference in texture space. In practice, they use distance in LAB space as a proxy for human perceptual distance. The distribution of transformation apply differently to 2D and 3D cases. In 2D, T includes rescaling, rotation, lightening, and additional noises. In 3D, camera distances, lighting conditions, translation and rotation, and texture difference are considered. The overall soft-constrained objective is defined as

$$\operatorname{argmax}_{x'} \mathbb{E}_{t \sim T} [\log P(y_t | t(x')) - \lambda \|LAB(t(x')) - LAB(t(X))\|] \quad (9)$$

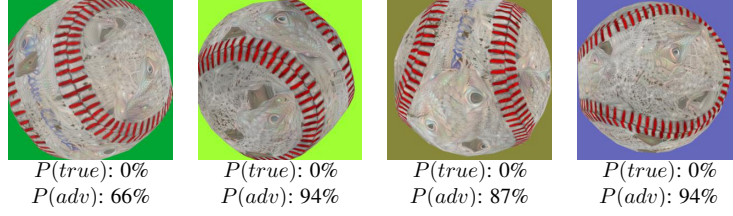


Figure 4: [1] The synthesized robust adversarial example to fool a network to misclassify a baseball ($P(\text{true})$) as a green lizard ($P(\text{adv})$). The attack is robust to lighting and geometric transformation.

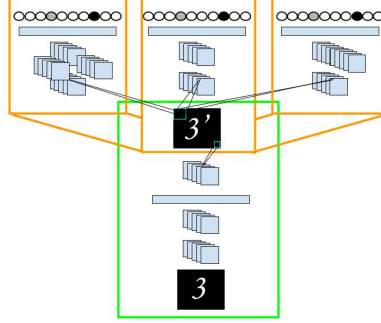


Figure 5: [2] ATN can be trained to predict adversarial examples. ATN takes input (clean image 3) and transform it into an adversarial example to a set of target networks, and fool them to predict 7.

105 4.2 Adversarial Transformation Networks

106 [2] propose an Adversarial Transformation Networks (ATN) to learn to generate adversarial examples.
 107 Similar to GAN, ATN is a neural network that transforms an input into an adversarial example against
 108 a target network or set of networks. Define ATN as

$$g_{f,\theta}(x) : x \in \mathcal{X} \rightarrow x' \quad (10)$$

where θ is the parameter vector of g , f is the target network which outputs a probability distribution across class labels, and $\text{argmax } f(x) \neq \text{argmax } f(x')$. The training objective is defined as

$$109 \quad \underset{\theta}{\text{argmin}} \sum_{x_i \in \mathcal{X}} \beta L_{\mathcal{X}}(g_{f,\theta}(x_i), x_i) + L_{\mathcal{Y}}(f(g_{f,\theta}(x_i)), f(x_i)) \quad (11)$$

110 where $L_{\mathcal{X}}$ is a loss function in the input space, $L_{\mathcal{Y}}$ is a specially-formed loss on the output space
 111 of f (described below) to avoid learning the identity function, and β is a weight to balance the two
 112 loss function. This is trained in a white-box fashion and requires no access to the network at test
 113 time. The transformed input can be targeted to particular wrong class t with a designed reranking
 114 function $r(y, t)$ for $L_{\mathcal{Y}}$ [2]. Adversarial example can be generated by Perturbation ATN (P-ATN) or
 115 Adversarial Autoencoding (AAE). In P-ATN, ATN acts as a variation on the residual block, while
 116 AAE attempts to reconstruct the original input subject to the perturbation constraint.

117 4.3 Transferability for Black Box Attacks

118 While ATN works as black-box attack at test time, it's only trained for a particular set of networks.
 119 So long as both models perform the same task, we would expect adversarial examples that affect
 120 one model often affect other models, even if different architectures or different training sets are
 121 used. Therefore, an attacker may train a substitute model, craft adversarial examples against the
 122 substitute, and transfer them to a victim model [13]. Since black box attacks only assume oracle calls
 123 to neural network outputs, adversary can even synthetically generate a dataset and labeled by the
 124 target DNN. After that, they can use local substitute to craft adversarial examples for the target DNN

Transferability of adversarial attacks.						
Network	Resnet				VGG	
Stat./Alg.	Baseline	Tune.	Reweight	Adv.	Tune.	Adv.
L_2 : Clean	80.76%	84.91%	86.97%	81.06%	82.88%	78.48%
L_2 : 0.1	76.82%	81.06%	71.36%	51.52%	74.09%	78.48%
L_∞ : Clean	80.76%	84.91%	86.97%	84.09%	82.88%	66.67%
L_∞ : 0.001	73.79%	81.36%	70.61%	55.91%	74.39%	66.82%

Table 2: We generate adversarial examples with L_2 and L_∞ norm constraints to adversarially trained ResNet. We then use those examples to test the transferability of adversarial attacks to our fossil classification tasks. It turns out that the generated examples do not transfer very well as most accuracy for other model weights and architecture do not decrease by much.

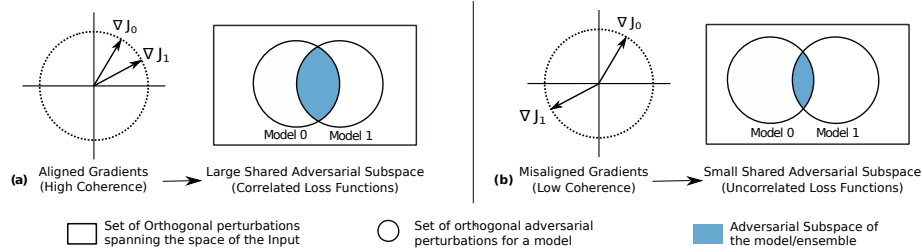


Figure 6: [9] Illustration showing the relationship between the gradient alignment and overlap of adversarial subspaces of two model. Diverse Training aims to train an ensemble with a smaller overlap in the adversarial subspace to avoid adversarial attacks by transfer.

[14]. [13] showed that machine learning approaches are in general vulnerable to systematic black-box attacks regardless of their structures, such as SVM, kNN, or DNN. They show effective attacks on standard model publicly available on Google and Amazon within 700 queries. In our experiments on transferability of the adversarial examples generated by PGD (shown in Table 2), adversaries for adversarially trained ResNet do not act as a large threat to other trained networks in our experiments.

4.4 Adversarial Subspace for Defense and Attacks

It has been shown that models with high dimensionality of adversarial subspace [20] are more likely to intersect, causing adversarial examples to transfer between models. It's natural to explore the attacks and defense of ensemble adversarial training because the attack vector must now fool multiple models in the ensemble instead of just a single model. Moreover, [19; 9] use experiments to show that ensemble of networks increases the diversity of perturbations seen during training and decouples adversarial example generation from the parameters of the trained model. Ensemble Adversarial Training augments a model's training data with adversarial examples crafted on other static pre-trained models, which relies on the transferability among models. [9] proposes gradient alignment loss (GAL) to train an ensemble of diverse models with misaligned gradients, thus reducing the dimensionality of the shared adversarial subspace. Intuitively, if the gradients of two models are misaligned, the perturbations that cause J_0 to increase do not cause J_1 to increase. [9] define the cosine similarity between gradients

$$CS(\nabla_x J_0, \nabla_x J_1) = \frac{\langle \nabla_x J_0, \nabla_x J_1 \rangle}{\|\nabla_x J_0\| \|\nabla_x J_1\|} \quad (12)$$

The GAL loss is defined as the smooth approximation to max operator

$$GAL = \log \left(\sum_{1 \leq a < b \leq N} \exp(CS(\nabla_x J_a, \nabla_x J_b)) \right) \quad (13)$$

They propose using the GAL cost as a regularization term for Diversity Training.

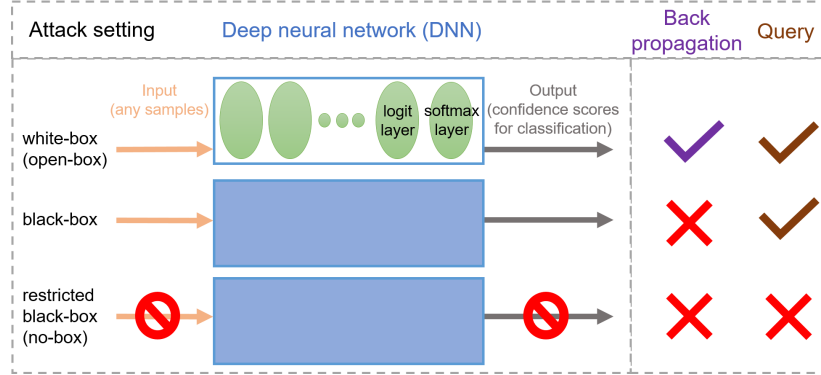


Figure 7: [3] Taxonomy of adversarial attacks to deep neural networks. Zeroth order optimization (middle) requires only query of the network.

145 4.5 Zeroth Order Attacks

146 To generate black-box adversarial attacks without transfer learning, several papers have proposed
 147 zeroth order optimization with only oracle calls. These methods [3; 12] often use numerical approxi-
 148 mations to compute higher order information such as gradients and Hessians, and achieve effective
 149 attacks on datasets such as CIFAR and MNIST. [3] directly estimate the gradients of the targeted DNN
 150 for generating adversarial examples. Specifically, their ZOO algorithm uses stochastic coordinate
 151 descent and effectively attack DNN and speed up computation time and reduce number of queries by
 152 attack-space dimension reduction, hierarchical attacks and importance sampling. They use numerical
 153 approximation to compute gradient and Hessian of a random coordinate, and apply different algorithm
 154 such as ADAM and Newton’s Method to minimize the softmax probability of the predicted class.
 155 They use dimension reduction techniques to reduce the number of estimated gradients and use a
 156 hierarchical attack scheme to gradually increase the number of applied transformation. They also
 157 propose to optimize the important pixels first, for instance, the pixels near the main object, by using
 158 importance sampling.

159 [7] instead propose a simple black-box attacks guided by output probabilities to search for adversarial
 160 images without transferability. Their proposed Simple Black-box Attack (SimBA) that takes as input
 161 the target image label pair (x, y) , a set of orthonormal candidate vectors Q and a step-size $\eta > 0$.
 162 Their algorithm is then simple as the intuition: for any direction $q \in Q$ and some step size η , one of
 163 $x + \eta q$ or $x - \eta q$ is likely to decrease $p(y|x)$. Their algorithm can be therefore implemented easily
 164 and demonstrated effective attacks to models without any extra information than oracle calls.

165 5 Conclusion

166 In this work, we evaluate two training methods, meta example reweighting and adversarial training,
 167 on fossil dataset to address corruptions at training time and test time. Moreover, we survey current
 168 empirical methods on generating black-box and white-box attacks to DNN, with a special focus
 169 on PGD robust to transformation, adversarial transformation network, transferability, adversarial
 170 subspace, and zeroth order attacks. These methods have shown effectiveness in defending and
 171 attacking neural networks in certain settings and are easy to implement and understand. As certified
 172 defense methods are developed to be more scalable [17; 4] and shown provable bounds, it would be
 173 interesting to compare and analyze those methods with empirical methods.

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